

Linea Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Consensys / Linea on Linea. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated May 24, 2024, was delivered to Consensys / Linea, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 15 days, the Cyfrin team conducted an audit on the Linea code provided by Consensys / Linea. The findings consist of 2 High, 2 Medium & 4 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Linea
Repository	zkevm-monorepo
Commit	364b64d2a270...
Audit Timeline	April 29th - May 17th, 2024
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	2	2	0
Medium Risk	2	0	2
Low Risk	4	3	1
Informational	6	2	4
Gas Optimizations	13	8	5
Total	27	15	12

6 Audit Scope

The audit scope was limited to Solidity smart contracts primarily covering the messaging service, token bridge and rollup, aiming to audit the following new developments:

- gas optimizations to blob submission and finalization for existing implementation of EIP-4844
- changes to L1 messaging and claiming of messages

All code relating to zk provers and other code outside the smart contract layer was out of scope. The following contracts were included in the scope for this audit:

```
contracts/LineaRollup.sol
contracts/ZkEvmV2.sol
contracts/interfaces/IGenericErrors.sol
contracts/interfaces/IMessageService.sol
contracts/interfaces/IPauseManager.sol
contracts/interfaces/IRateLimiter.sol
contracts/interfaces/l1/IL1MessageManager.sol
contracts/interfaces/l1/IL1MessageManagerV1.sol
contracts/interfaces/l1/IL1MessageService.sol
contracts/interfaces/l1/ILineaRollup.sol
contracts/interfaces/l1/IZkEvmV2.sol
contracts/interfaces/l2/IL2MessageManager.sol
contracts/interfaces/l2/IL2MessageManagerV1.sol
contracts/lib/Utils.sol
contracts/messageService/MessageServiceBase.sol
contracts/messageService/l1/L1MessageManager.sol
contracts/messageService/l1/L1MessageService.sol
contracts/messageService/l1/TransientStorageReentrancyGuardUpgradeable.sol
contracts/messageService/l1/v1/L1MessageManagerV1.sol
contracts/messageService/l1/v1/L1MessageServiceV1.sol
contracts/messageService/l2/L2MessageManager.sol
contracts/messageService/l2/L2MessageService.sol
contracts/messageService/l2/v1/L2MessageManagerV1.sol
contracts/messageService/l2/v1/L2MessageServiceV1.sol
contracts/messageService/lib/PauseManager.sol
contracts/messageService/lib/RateLimiter.sol
contracts/messageService/lib/SparseMerkleTreeVerifier.sol
contracts/messageService/lib/TimeLock.sol
contracts/messageService/lib/TransientStorageHelpers.sol
contracts/tokenBridge/BridgedToken.sol
contracts/tokenBridge/TokenBridge.sol
contracts/tokenBridge/interfaces/ITokenBridge.sol
```

At the conclusion of the audit the verified changes were merged into the linea-contracts repository at commit b17e7c7.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Linea is a Type 2 zkEVM L2 rollup with full EVM equivalence that aims to provide the “Ethereum experience” at scale.

9 Attestation

Cyfrin attests that it performed a security review of the Linea code provided by Consensys / Linea.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin’s repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-05-24-cyfrin-linea-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin’s full audit report with all findings provided to Consensys / Linea has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=1016716cf00dab3448e645c882fff9ae9e4f69f3b3d378438566dfc9e16d4c2b
F=2024-05-24-cyfrin-linea-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.